Roll No.: DL.AI.U4AID24140

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Amrita School of Artificial Intelligence, Faridabad

B.Tech Mid-Term Examinations – October 2024

First Semester

Artificial Intelligence

23AID102 Elements of Computing - 1

Duration: Two hours

Maximum: 50 Marks

Course Outcomes (COs):

CO	Course Outcomes
CO1	Demonstrate proficiency in performing number system conversion, manipulating Boolean Algebra expressions and realization of basic gates .
CO2	Demonstrate proficiency in performing number system conversion, manipulating Boolean Algebra expressions and realization of basic gates ..
CO3	Understand instruction set architecture and computer system components.

Answer all questions

Question No-1 Multiple choice questions **[5x2][CO1][BTL 2]**

Q. 1A. What is the result of adding the binary numbers 1101 and 1011?

a) 11000 b) 10100 c) 11010 d) 11100

Q. 1B. Convert the decimal number 315 to its hexadecimal equivalent.

a) 14B b) 13C c) 13B d) 12B

Q. 1C. In a 4-bit signed magnitude, what is the decimal equivalent of the binary number 1011?

a) -3 b) -7 c) 7 d) -11

Q. 1D. Which logic gate is known as the universal gate?

a) AND b) OR c) NAND d) XOR

Q. 1E. A multiplexer selects one input out of how many possible inputs?

a) 2^n b) n c) 2n d) 4n

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Question No-2. Convert the decimal number 275 to its binary, octal, and hexadecimal equivalents [6][CO1][BTL2]

Question No-3. Subtract 1010 (which is +10 in decimal) from 0110 (which is +6 in decimal) using 4-bit 2's complement representation. [6][CO1][BTL2]

Question No-4. Design a 4-to-1 multiplexer has 4 input lines (I0, I1, I2, I3), 2 select lines (S0, S1), and 1 output (Y). [6][CO1][BTL2]

Question No-5. Optimize the Boolean expression F (A,B,C) for the provided truth table as below using K-map. [6][CO1][BTL2]

A	B	C	F
0	0	0	1
0	0	1	0
0	1	0	1
0	1	1	X
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	X

Question No-6. Explain the concept of adding bits in a stream of binary number (e.g., 1001+0001) on hardware implementation with a help of full adder circuit and a suitable diagram.

[8][CO1][BTL3]

Question No-7. Explain the different methods of representing signed numbers in binary with suitable examples? [8][CO1][BTL3].

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Course Outcome /Bloom's Taxonomy Level (BTL) Mark Distribution Table

CO	Marks	BTL	Marks
CO1	**50**	BTL 2	**34**
		BTL 3	**16**

Bloom's Taxonomy Levels (attached for reference)

Level 1 – Remember

Level 2 - Understand

Level 3 – Apply

Level 4 - Analyze

Level 5 - Evaluate

Level 6- Create

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